

a

1 Per-layer bytes

int4 KV = 528 B / token /
GQA layer

fp32 / reference

fp32 temporal + bf16
conv
= 1,085,440 B / GDN layer

bf16

bf16 temporal + bf16
conv
= 561,152 B / GDN layer

recurrent bytes
-48.30%
(not exactly half)

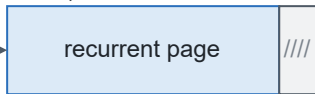
2 Padded recurrent page

$B = \text{padded recurrent-page bytes} / 528 \text{ B per token}$

fp32 / reference

1,089,792 B

$B = 2,064 \text{ tokens}$

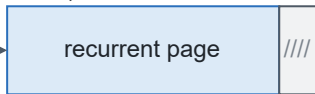


padding / alignment

bf16

566,016 B

$B = 1,072 \text{ tokens}$

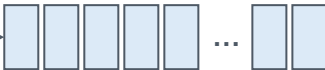


padding / alignment

3 Shared fixed GPU pool

fp32 / reference

$K = 3,287 \text{ allocated blocks}$



bf16

$K = 6,330 \text{ allocated blocks}$

4 Discrete capacity at $L = 4,096$

$S_{\text{alloc}} = K / (H + \text{ceil}(L/B)), H = 3$

fp32 / reference

$3,287 / (3 + 2) = 657.4$
allocator-equivalent slots

bf16

$6,330 / (3 + 4) = 904.3$
allocator-equivalent slots

These are allocator-equivalent slots,
not concurrent requests.

No scheduler behavior, request admission, throughput, or latency is implied.